

Pascal's triangle



1 The 4th row of Pascal's triangle consists of the numbers 1, 4, 6, 4, 1.

- a Write down the numbers in the 5th row of Pascal's triangle.
- b Find the sum of the entries in the 5th row of Pascal's triangle.

2 Complete the following entries in a particular row of Pascal's triangle:

$$1, 8, \mathbb{I}, 56, 70, \mathbb{I}, 28, \mathbb{I}, 1$$

3 Which row of Pascal's triangle would give us the coefficients of terms in the expansion of $(x + y)^7$?

4 State the number of terms in the following expansions:

a $(4x + y)^6$

b $(m + y)^8$

5 Consider the expansion of $(y + k)^4$.

- a How many terms will there be in the expansion?
- b Write the coefficient of the 2nd term.

6 Using the relevant row of Pascal's triangle, determine the coefficient of each term in the following expansion:

$$(x + m)^4 = \mathbb{I}x^4m^0 + \mathbb{I}x^3m^1 + \mathbb{I}x^2m^2 + \mathbb{I}x^1m^3 + \mathbb{I}x^0m^4$$

Binomial theorem

7 Evaluate the following:

a $\binom{6}{5}$

b $\binom{14}{1}$

c $\binom{10}{9}$

d $\binom{8}{0}$

8 Consider the expansion of $(a - 4)^5$. Will the 2nd term be positive or negative?

9 By considering $(x + y)^4 = (x + y)(x + y)(x + y)(x + y)$, in how many ways can three factors of x and one factor of y be chosen from the expansion to form the term which contains x^3y^1 ?

10 Consider the binomial $(4x + 3y)^4$.



11 Find the specified term for the following expansions:

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|--|---|
| a Fifth term of $(b + 2)^7$. | b Fifth term of $(c + d)^9$. |
| c Fifth term of $(3u - v)^6$. | d Tenth term of $\left(u + \frac{1}{2}\right)^{12}$. |
| e Seventh term of $\left(\frac{3y}{2} - \frac{2}{3y}\right)^{10}$. | f Eleventh term of $(2x + y^2)^{13}$. |
| g Sixteenth term of $(x - y^5)^{19}$. | h Middle term of $(4x^2 + 3y^3)^8$. |
| i Middle term of $(-3x^{-3} + 2y^{-2})^4$. | j Constant term of $\left(2y - \frac{1}{y^3}\right)^8$. |

12 The first three terms of $(x + y)^{10}$ are x^{10} , $10x^9y$ and $45x^8y^2$. Using the symmetry of the coefficients, write the last three terms of the expansion.

13 In the expansion of $(m + k)^n$, the coefficient of the 9th term is 45.

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|----------------------------------|-----------------------------|
| a Find the value of n . | b Find the 9th term. |
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14 A particular term in the expansion of $\left(3a^2 + \frac{p}{b}\right)^4$ is $\frac{96a^2}{b^3}$, for some constant p . Find the value of p .

15 Consider the expansion of $(a + b)^5$.

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| a Find the coefficient of the 3rd term. |
| b Which other term has the same coefficient as the 3rd term? |

16 Which two terms in the expansion of $(u + v)^{11}$ have a coefficient of $\binom{11}{9}$?

17 Find the coefficient of x^{17} in the expansion of $(3x^2 + 2x)^{11}$.

18 Use the binomial theorem to expand the following expressions:

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|---|---|------------------------|---|
| a $(1 + x)^4$ | b $(2 + b)^5$ | c $(p + q)^5$ | d $(c - d)^8$ |
| e $(u + 4v)^6$ | f $(y + 3)^4$ | g $(ka - b)^6$ | h $(y - 5)^3$ |
| i $\left(y - \frac{1}{3}\right)^3$ | j $\left(2 + \frac{y}{2}\right)^3$ | k $(3y + 2r)^3$ | l $\left(a - \frac{1}{a}\right)^3$ |
| m $(u^2 + 3v^2)^3$ | n $\left(1 + \frac{4}{y}\right)^3$ | | |

19 Find:



- a The term in $(3 + \sqrt{x})^{10}$ that contains x^4 . b The term in $(c + d)^{10}$ that contains c^3 .
c The term in $(u + v)^{11}$ that contains v^8 . d The term in $(u + v)^{15}$ that contains u^5 .

20 If the eighth and tenth terms in the expansion of $(x + y)^n$ have the same coefficients, find the value of n .

21 Find the coefficient of a^3b^6 in the expansion of $\left(2a - \frac{b}{2}\right)^9$.

22 Consider the expansion of $(1 + x)^9$. State the coefficient of the 4th term.

23 Find the value of n such that the expansion of $(a + b)^n$ contains the term $84a^6b^3$.

24 Consider the binomial series:

$$(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

- a Use the first four terms to approximate $(1.03)^{-2}$ to the nearest thousandth.
b Use the first four terms to approximate $(1.05)^{\frac{3}{4}}$ to the nearest thousandth.
c Use the first five terms to approximate $\frac{1}{(1.06)^2}$ to the nearest thousandth.